

2017-Ph-H1-Q23

Section: Electricity

Topic: Current, PD, Power, Resistance

A current flows through a resistor for a time.

What is the energy transferred?

 **Use:**

$$E = IVt$$

Given:

- $I = 0.25 \text{ A}$
- $V = 6.0 \text{ V}$
- $t = 300 \text{ s}$

$$E = 0.25 \times 6.0 \times 300 = 450 \text{ J}$$

 **Final Answer: A — 450 J**

 **Revision Tips:**

- Also valid:

$$E = Pt = I^2Rt$$